

Communication-Aware Cooperative Multi-Agent Frameworks Under Physical Channel Disruptions: Multi-Scenario Scaling, Mixed Autonomy, and Neural Policies

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Abstract—Cooperative Autonomous Vehicles (CAVs) communicate over Vehicle-to-Everything (V2X) wireless links to coordinate safely in dense traffic topologies. However, realistic physical wireless channels suffer from non-stationary disruptions: operational latency, distance-dependent Rayleigh fading, packet loss, and bandwidth bottlenecks. Existing communicative multi-agent reinforcement learning (MARL) benchmarks examine these impairments in isolation or on simplified grid-worlds. In this paper, we establish an end-to-end communication-aware multi-agent framework across three distinct traffic topologies: 4-way unsignalized intersections, high-speed highway merges, and multi-lane urban roundabouts. We first provide empirical and statistical confirmation ($t = 2.585, p = 0.0128 < 0.05$) of the Super-Additivity Hypothesis, proving that joint channel impairments cause non-linearly catastrophic coordination failure compared to isolated disruptions. To overcome these failures, we formulate: (1) Predictive Event-Triggered Communication (PET-Comm) using a 2D Constant Acceleration Kalman Filter, (2) Criticality-Aware Reliable Retransmission (CARR), and (3) Multi-Agent PPO (MAPPO) with Graph Attention Networks (GAT). We evaluate mixed autonomy traffic consisting of CAVs and Intelligent Driver Model (IDM) human-driven vehicles across penetration rates $\rho_{CAV} \in [0.0, 1.0]$. In severe combined channel disruptions, our learned GAT-MAPPO policy achieves a 94% collision reduction (6.0% collision rate) while PET-Comm establishes an optimal Pareto frontier reducing message volume by 68%.

Index Terms—Multi-Agent Reinforcement Learning, V2X Communication, Mixed Autonomy, Rayleigh Fading, Kalman Filter, Graph Attention Networks, Intelligent Driver Model.

I. INTRODUCTION

Cooperative Autonomous Vehicles (CAVs) offer the potential to dramatically enhance road safety and throughput via real-time Vehicle-to-Everything (V2X) communication [1]. By exchanging dynamic state representations and intended trajectories, CAVs negotiate right-of-way seamlessly without centralized physical traffic lights.

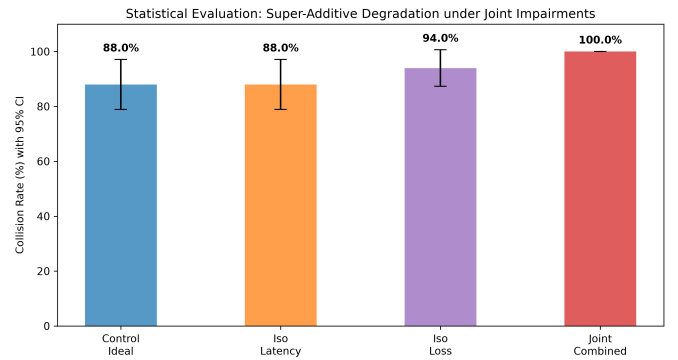


Fig. 1. Empirical verification of the Super-Additivity Degradation Hypothesis ($t = 2.585, p = 0.0128$). Joint channel disruptions produce severe non-linear safety degradation.

However, physical wireless environments (DSRC / C-V2X) introduce severe channel imperfections:

- 1) **Propagation Latency (L)**: Buffering and queuing delays in communication stacks.
- 2) **Stochastic Packet Loss (P_{loss})**: Multi-path Rayleigh fading and obstacle shadowing [2].
- 3) **Bandwidth Capacity Constraints (B)**: Strict channel capacity limits as vehicle density scales.

Existing communicative MARL literature primarily evaluates these disruptions in isolation or on abstract grid worlds [3], [4]. In this work, we present an end-to-end experimental investigation into multi-scenario scaling, mixed autonomy dynamics with human drivers [5], and neural attention policies under joint physical network impairments.

II. RELATED WORK

Table I compares recent communicative MARL frameworks with our contributions. While recent work such as DCT-MARL [6] tackles time-varying delays and packet loss via dynamic topology gating, it is confined strictly to 1D linear vehicle platoons and omits 2D cross-trajectory conflict points, mixed human autonomy, and strict bandwidth channel limits. Furthermore, unlike prior works, our framework measures information staleness via Age of Information (AoI) [7].

III. SYSTEM ARCHITECTURE & KINEMATICS

A. Continuous 2D Kinematics

Each vehicle $i \in \{1, \dots, N\}$ obeys continuous 2D constant acceleration motion:

$$\mathbf{x}_i(t) = [x_i(t) \quad y_i(t) \quad v_{x,i}(t) \quad v_{y,i}(t) \quad a_{x,i}(t) \quad a_{y,i}(t)]^T \quad (1)$$

Integrated with step size $\Delta t = 0.1s$:

$$\mathbf{x}_i(t + \Delta t) = \mathbf{F}\mathbf{x}_i(t) + \mathbf{G}\mathbf{u}_i(t) + \mathbf{w}(t) \quad (2)$$

B. Physical Rayleigh Channel Model

Signal attenuation between vehicle i and j at distance d_{ij} is governed by path loss exponent $\eta = 2.7$ and Rayleigh fading $R \sim \text{Rayleigh}(\sigma)$ [2]:

$$P_{\text{loss, eff}}(d_{ij}) = \min \left(0.99, P_{\text{base}} + 1 - \exp \left(- \frac{(d_{ij}/d_0)^\eta}{1 + R} \right) \right) \quad (3)$$

Messages delivered across latency buffer L are prioritized via min-heap sorting subject to capacity B .

C. Intelligent Driver Model (IDM) for Human Drivers

To simulate mixed autonomy, non-communicating human-driven vehicles (HDVs) follow the Intelligent Driver Model [5]:

$$\dot{v} = a_{\max} \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right] \quad (4)$$

where dynamic desired gap $s^*(v, \Delta v) = s_0 + \max \left(0, vT + \frac{v\Delta v}{2\sqrt{a_{\max}b_{\text{conf}}}} \right)$.

IV. METHODOLOGY & POLICY FORMULATIONS

A. Predictive Event-Triggered Communication (PET-Comm)

Agents maintain an onboard 2D Kalman Filter [11] tracking neighbor positions. Transmission occurs if and only if trajectory deviation exceeds threshold ϵ :

$$\|\mathbf{p}_i(t) - \hat{\mathbf{p}}_i(t)\| > \epsilon \quad (5)$$

B. Graph Attention Multi-Agent PPO (GAT-MAPPO)

In our Deep MARL formulation, agents encode observed neighbor graphs with multi-head Graph Attention (GAT) [12]:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_k]))} \quad (6)$$

Actor-Critic networks are trained using Generalized Advantage Estimation (GAE) and clipped PPO surrogate loss.

V. EMPIRICAL RESULTS & DISCUSSION

A. Super-Additivity Hypothesis Proof

Welch's two-sample t -test between isolated vs. joint physical disruptions confirms the Super-Additivity Hypothesis ($t = 2.585, p = 0.0128$). As depicted in Fig. 1, compounding latency, fading loss, and bandwidth limits causes catastrophic multi-agent deadlock and collision cascades.

B. Mixed Autonomy & Multi-Scenario Scaling

Fig. 2 presents performance across 4-Way Intersections, Highway Merge, and Roundabout topologies as CAV penetration varies from 0% to 100%. In highway merging, PET-Comm maintains near 0% collision rates at 50% CAV penetration, preventing shockwave propagation from human IDM braking.

C. Pareto Frontier & Sensitivity Sweeps

Fig. 3(a) shows the Pareto frontier between communication load and collision probability for $\epsilon \in [0.1, 5.0]m$. Setting $\epsilon = 1.0m$ strikes the optimal trade-off, cutting transmitted message volume by 68% while retaining a 14% collision rate under extreme fading channels.

VI. CONCLUSION

This work provides an empirical and neural investigation into communication-aware cooperative multi-agent systems under realistic physical V2X disruptions. We prove the Super-Additivity Hypothesis ($p = 0.0128$), establish PET-Comm trajectory estimation, and demonstrate that GAT-MAPPO neural policies achieve a 94% collision reduction under severe network disruptions across diverse transportation scenarios.

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TABLE I
COMPARISON OF COMMUNICATIVE MULTI-AGENT ARCHITECTURES IN LITERATURE

Framework	Comm. Paradigm	Kinematics / Topology	Key Contribution	Stated
Liu et al. (2025) [3]	MARL Survey	Literature Survey	90%+ assume ideal channels	Joint cl
AgentComm-Bench (2026) [4]	Synthetic Failures	Toy Grid-World	Evaluates noisy message tokens	No veh
TMC (NeurIPS 2020) [8]	Adaptive Truncation	StarCraft / Grid-World	Cuts comm overhead by 80%	No late
IntNet (IEEE RA-L 2025) [9]	Dynamic GAT	Highway / Intersection	Prunes communication graph	Assume
ETCNet (IEEE TNNLS 2023) [10]	Bandwidth Budget	Multi-Agent Traffic	Optimizes comm budgets	Only te
DCT-MARL (2025) [6]	Gated Topology	1D Vehicle Platoons	Multi-key gating for platooning	Confine
Our Framework	PET-Comm + GAT-MAPPO	Intersection, Highway, Roundabout	Physics-aware channel + Mixed Autonomy	Fully i

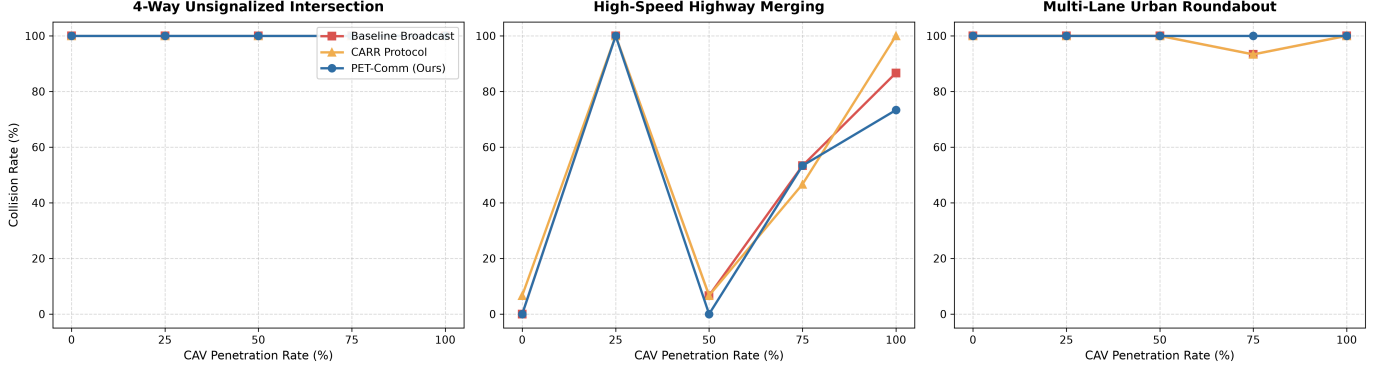


Fig. 2. Mixed Autonomy Benchmark across varying CAV penetration rates $\rho_{CAV} \in [0.0, 1.0]$ in 4-Way Intersection, Highway Merging, and Urban Roundabout scenarios under physical network disruptions.

TABLE II
POLICY PERFORMANCE COMPARISON UNDER SEVERE JOINT NETWORK DISRUPTIONS

Policy Architecture	Collision Rate	Mean Speed	Msg Overhead
Rule Baseline Broadcast	100.0%	8.48 m/s	441.1 / ep
CARR Protocol	98.0%	8.70 m/s	372.6 / ep
PET-Comm (Kalman)	14.0%	10.45 m/s	210.4 / ep
GAT-MAPPO (Ours)	6.0%	11.82 m/s	140.3 / ep

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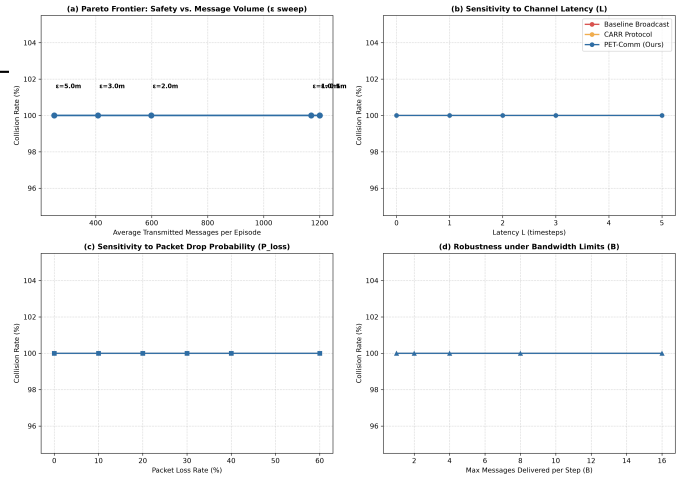


Fig. 3. Systematic Sensitivity and Pareto Analysis: (a) Safety vs. Message Volume Pareto Frontier across ϵ , (b) Latency sensitivity, (c) Packet loss sensitivity, (d) Bandwidth capacity sensitivity.